1 Genome sequencing1.1 Extraction of genome DNA

Genomic DNA was extracted with the SDS method[1]. The harvested DNA was detected by the agarose gel electrophoresis and quantified by Qubit® 2.0 Fluorometer (Thermo Scientific).

1.2 Library construction

A total amount of 1μg DNA per sample was used as input material for the DNA sample preparations. Sequencing libraries were generated using NEBNext® Ultra™ DNA Library Prep Kit for Illumina (NEB, USA) following manufacturer's recommendations and index codes were added to attribute sequences to each sample. Briefly, the DNA sample was fragmented by sonication to a size of 350bp, then DNA fragments were end-polished, A-tailed, and ligated with the full-length adaptor for Illumina sequencing with further PCR amplification. At last, PCR products were purified (AMPure XP system) and libraries were analysed for size distribution by Agilent2100 Bioanalyzer and quantified using real-time PCR.

1.3 Sequencing

The whole genome of XXX was sequenced using Illumina NovaSeq PE150 at the Beijing Novogene Bioinformatics Technology Co., Ltd.

2 Genome assembly

2.1 Data processing

The raw data obtained by sequencing (Raw Data) had a certain proportion of low-quality data. In order to ensure the accuracy and reliability of the subsequent information analysis results, the original data must be filtered to obtain valid data (Clean Data).

The specific processing steps included as follows:

1)The reads containing low-quality bases (mass value ≤20) over a certain percentage (the default was 40%) were removed;

2) The number of N in reads beyond a certain proportion (the default was 10%) were removed;

3) Some reads, the overlap between them and the Adapter, which exceeded a certain threshold (the default was 15bp) and had less then 3 mismatches between them, were removed;

4) If the sample such as minigenome was contaminated by host, it needs to be blast with the host data to filter out reads that may originate from the host.

2.2 Assembly

The specific processing steps for genome assembly with Clean Data included as follows:

1) Assembled with SOAP denovo software:

Different K-mers (the default were 95, 107, 119) were selected for assembly, According to the project type, the optimal K-mer, further adjusting other parameters (-d -u -R - F, etc.) and the least scaffolds were choosed as the preliminary assembly result;

2) Assembled with SPAdes software:

Different K-mers (the default were 99 and 127) were selected for assembly, According to the project type, the assembly result was obtained with the optimal kmer and the least scaffolds;

3) Assembled with Abyss software:

K-mer 64 was selected for assembly and the assembly result was obtained:

4) The assembly results of the three softwares were integrated with CISA software and the assembly result with the least scaffolds was selected;

5)The gapclose software was used to fill the gap of preliminary assembly results.The same lane pollution by filtering the reads with low sequencing depth (less than 0.35 of the average depth) was removed to obtain the final assembly result;

6) Fragments below 500 bp were filtered out and the final result was counted for gene prediction.

3 Genome Component prediction

Genome component prediction included the prediction of the coding gene, repetitive sequences, non-coding RNA, genomics islands, transposon, prophage, and clustered regularly interspaced short palindromic repeat sequences (CRISPR). The available steps were proceeded as follows:

1) For bacteria, we used the GeneMarkS[6] program to retrieve the related coding gene.

2) The interspersed repetitive sequences were predicted using the RepeatMasker[7] (http://www.repeatmasker.org/). The tandem Repeats were analyzed by the TRF[8] (Tandem repeats finder).

3) Transfer RNA (tRNA) genes were predicted by the[9][10].tRNAscan-SE. Ribosome RNA (rRNA) genes were analyzed by the rRNAmmer Smallnuclear RNAs(snRNA)were predicted by BLAST against the Rfam[11][12]database.

4) The[13]IslandPath-DIOMB program was used to predict the Genomics Islands, and transposonPSI was used to predict the transposons based on the homologous blast method. The[14]PHAST was used for the prophage prediction(http://phast.wishartlab.com/) and the[15]CRISPRFinder was used for the CRISPR identification.

4 Gene function

We used seven databases to predict gene functions. They were respective GO[16] (Gene Ontology), KEGG[17][18] (Kyoto Encyclopedia of Genes and Genomes), COG[19](Clusters of Orthologous Groups), NR[20](Non-Redundant Protein Database), TCDB[21](Transporter Classification Database), and, Swiss-Prot[22]. A whole genome Blast[23] search (E-value less than 1e-5, minimal alignment length percentage larger than 40%) was performed against above seven databases. The secretory proteins were predicted by the SignalP[24]database, and the prediction of Type I-VII proteins secreted by the pathogenic bacteria were based on the EffectiveT3[25] software. Meanwhile, we analyzed the secondary metabolism gene clusters by the antiSMASH[26].For pathogenic bacteria, we added the pathogenicity and drug resistance analyses. We used the PHI[27](Pathogen Host Interactions),VFDB[28](Virulence Factors of Pathogenic Bacteria), ARDB[29](Antibiotic Resistance Genes Database) to perform the above analyses. Carbohydrate-Active enzymes were predicted by the Carbohydrate-Active enZYmesDatabase[30].

5 Comparative genomics analysis

Comparative genomic analysis included the genomic synteny, the core genes and specific genes, gene family phylogenetic tree, SNP(Single Nucleotide Polymorphism), indel (insertion and deletion) and SV(Structural Variation) annotation, and genome visualization.
1)Genomic alignment between the sample genome and reference genome (or among more than two sample genomes) were performed using the MUMmer [31] and LASTZ[32][33] tools. Genomic synteny was analyzed based on the alignment results.
2) Core genes and specific genes were analyzed by the CD-HIT rapid clustering of similar proteins software with a threshold of 50% pairwise identity and 0.7 length difference cutoff in amino acid. Then the venn figure was drawn to show their relationships among the samples.
3) Gene family was constructed by the gene of XXX and XXX, using the multiple softwares: Blast[34] was used to pairwise align all genes and eliminate the redundancy by solar and carried out gene family clustering treatment based on the alignment results with Hcluster_sg software.
4) The phylogenetic tree was constructed by the TreeBeST or PhyML and the setting of bootstraps was 1,000 with the orthologousgenes.
5) SNP, indel and SV were found by the genomic align ment results among samples by the MUMmer and LASTZ we mentioned them before.

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